

- 1 (a) Find the set of values of the constant k for which the line $y + k = x$ does not intersect the curve $y = kx^2 + 7x$. [3]

- (b) Given that the equation of the curve is $y = 4x^2 + p(x + 1) - 3$, where p is a constant. Find the values of p for which the x -axis is a tangent to the curve. [3]

2 (a) Factorise $a^3 - b^3$.

[1]

(b) Hence solve the equation $x^3 - (x - 6)^3 = 504$.

[4]

3 Express $\frac{2x-5}{(2x+1)(4x^2-1)}$ in partial fractions.

[5]

- 4 (a) Variables x and y are related by the equation $y = \frac{3x^2}{2-x^3}$. When a graph of $\frac{y}{x^2}$ is plotted against xy , a straight line is obtained. Given further that the point $(k, 5)$ lies on the straight line, find the value of k . [3]

- (b) Variables x and y are related by the equation $x^{p+1}y^q = e$, where p and q are constants. Explain clearly how a straight line graph can be drawn to represent this equation. You should state the variables plotted on each axis, and express the gradient and the vertical-intercept of this straight line drawn in terms of p and/ or q . [3]

- 5 The curve $y = \frac{1}{(x^2 - kx)^3}$, where k is a constant, has a stationary point at $x = 2$.
Find the coordinates of this stationary point and determine its nature.

[6]

6 Given that $y = 3x^n \ln(4x)$, where n is a constant.

(a) Find $\frac{dy}{dx}$. [2]

(b) Hence, find $\int x^{n-1} \ln(4x) \, dx$. [4]

7 (a) Prove that $\operatorname{cosec} 2x \sec 2x = \tan 2x + \cot 2x$.

[3]

(b) Hence solve the equation $\operatorname{cosec} 2x \sec 2x = 2$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.

[3]

- 8 Mr Lin bought a new electric car. The value of the car, in dollars, after t years, is given by $V = 15\,000 + 150\,388 e^{-kt}$, where k is a constant. The value of the car after 3 years is expected to be \$127 667.

Calculate

- (a) the initial value of the car, [1]

- (b) the value of the car, to the nearest dollars, after 5 years, [4]

- (c) the number of years, to the nearest years, when the car's expected value will be first less than \$30 000. [2]

- 9 The equation of a curve is $y = \frac{\sin^2 x}{2 - \cos 2x}$. Find the equation of the normal to the curve at the point $x = \frac{\pi}{6}$. Express your answer in the form $y = -a\sqrt{3}x + b\sqrt{3}\pi + c$ where a , b and c are constants. [7]

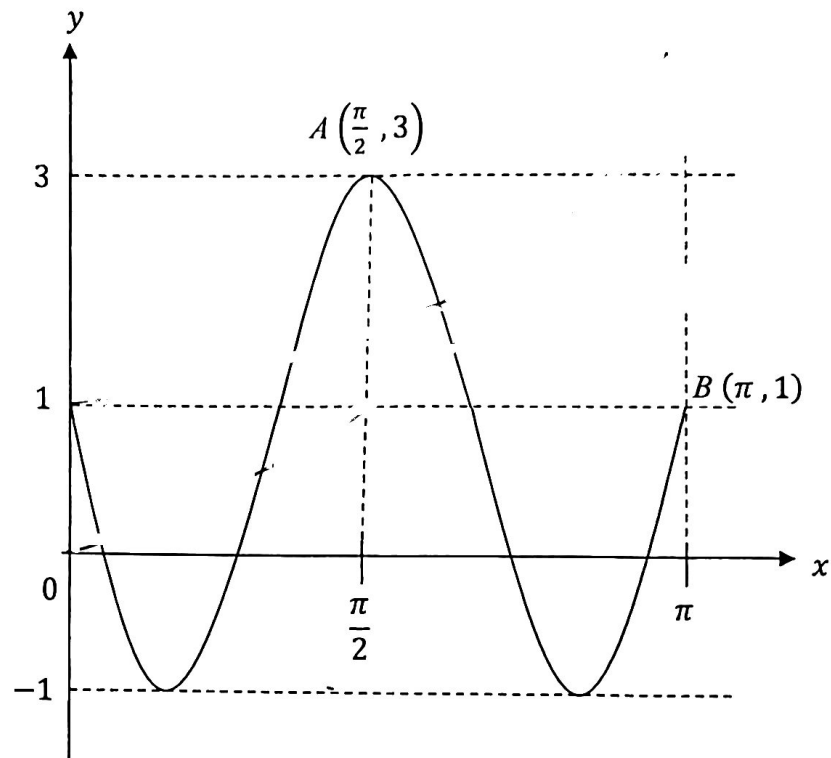
- 10 A circle has equation $x^2 + y^2 + 10x + py = q$, where p and q are constants. Given that the coordinates of the centre of the circle is $(s, 12)$ and the radius is 14 units.

(a) Find the values of p , q and s . [4]

(b) Write down the equations of the two tangents to the circle where they are parallel to the y -axis. [2]

(c) Mrs Xie says that it is possible to draw a tangent to the circle such that this tangent passes through the point $(-14, 4)$. Do you agree with her? Justify your answer. [2]

- 11 The diagram shows part of the curve $y = a \sin kx + b$, where a , b and k are constants. The coordinates of point A is $(\frac{\pi}{2}, 3)$ and point B is $(\pi, 1)$.



- (a) Find the values of a , b and k .

[4]

- (b) On the given diagram provided, sketch the graph of $y = \tan \frac{1}{2}x$ for $0 \leq x \leq \pi$.

Hence state the number of solutions to the equation $\tan \frac{1}{2}x = a \sin kx + b$ for
[2]

- (c) A line with the equation $y = mx + 3$ intersects the curve $y = a \sin kx + b$ at exactly 2 points for $0 \leq x \leq \frac{2\pi}{3}$. Find the set of values of m . [2]

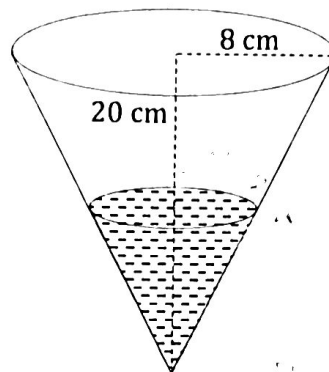
- 12 (a)(i) Write down and simplify the first three terms in the expansion, in ascending powers of x , of $(1 + px)^6$. [1]

- (ii) Given that the first two non-zero terms in the expansion, in ascending powers of x , of $(1 + qx)(1 + px)^6$ are 1 and $-\frac{21x^2}{4}$, find the possible values of p and q . [4]

(b)(i) If the fourth term in the binomial expansion of $\left(ax + \frac{b}{x}\right)^n$ is independent of x , find the value of n . [2]

(ii) Hence find the value of $\frac{a}{b}$ if the coefficient of second term is 50 times the coefficient of fifth term. [3]

- 13 A container is in the shape of an inverted right circular cone with radius 8 cm and height 20 cm. Water is poured into the container at a constant rate of $100 \text{ cm}^3/\text{s}$.



- (a) Given that h is the depth of the water, show that the volume of the water, $V \text{ cm}^3$, in the container is $V = \frac{4}{75}\pi h^3$. [2]

(b) Calculate, at the instant when the height of the water is 8 cm, the rate of increase of

(i) the depth of the water, [3]

(ii) the surface area of the container which is in contact with the water. [5]